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In [ ]: Question4 (B): Perform Basic Data Analysis on Product Inventory Using  
Pandas  
Problem Statement:  
You are given a dictionary containing information about products in an inventory, including their names, prices, and quantities available:  
data = {'Product': ['Laptop', 'Smartphone', 'Tablet', 'Monitor', 'Keyboard'], 'Price': [1200, 800, 300, 150, 50], 'Quantity': [10, 20, 15, 25, 50]} and do any five basic operations and plot.

In [1]: import pandas as pd  
import matplotlib.pyplot as plt  
  
data = {  
 'Product': ['Laptop', 'Smartphone', 'Tablet', 'Monitor', 'Keyboard'],  
 'Price': [1200, 800, 300, 150, 50],  
 'Quantity': [10, 20, 15, 25, 50]  
}  
  
df = pd.DataFrame(data)  
  
df['Total\_Value'] = df['Price'] \* df['Quantity']  
stats = df.describe()  
  
expensive\_products = df[df['Price'] > 200]  
  
sorted\_df = df.sort\_values(by='Quantity', ascending=False)  
  
top\_product = df.loc[df['Total\_Value'].idxmax()]  
  
print("--- Full Inventory DataFrame ---")  
print(df)  
print("\n--- Descriptive Statistics ---")  
print(stats)  
print("\n--- Products Price > 200 ---")  
print(expensive\_products)  
print("\n--- Top Product by Inventory Value ---")  
print(top\_product)  
  
plt.figure(figsize=(10, 6))  
plt.bar(df['Product'], df['Total\_Value'], color='skyblue')  
plt.xlabel('Product Name')  
plt.ylabel('Total Inventory Value (\$)')  
plt.title('Total Inventory Value per Product')  
plt.show()

--- Full Inventory DataFrame ---

|   | Product    | Price | Quantity | Total_Value |
|---|------------|-------|----------|-------------|
| 0 | Laptop     | 1200  | 10       | 12000       |
| 1 | Smartphone | 800   | 20       | 16000       |
| 2 | Tablet     | 300   | 15       | 4500        |
| 3 | Monitor    | 150   | 25       | 3750        |
| 4 | Keyboard   | 50    | 50       | 2500        |

--- Descriptive Statistics ---

|       | Price       | Quantity  | Total_Value  |
|-------|-------------|-----------|--------------|
| count | 5.000000    | 5.000000  | 5.000000     |
| mean  | 500.000000  | 24.000000 | 7750.000000  |
| std   | 486.055552  | 15.572412 | 5921.359641  |
| min   | 50.000000   | 10.000000 | 2500.000000  |
| 25%   | 150.000000  | 15.000000 | 3750.000000  |
| 50%   | 300.000000  | 20.000000 | 4500.000000  |
| 75%   | 800.000000  | 25.000000 | 12000.000000 |
| max   | 1200.000000 | 50.000000 | 16000.000000 |

--- Products Price > 200 ---

|   | Product    | Price | Quantity | Total_Value |
|---|------------|-------|----------|-------------|
| 0 | Laptop     | 1200  | 10       | 12000       |
| 1 | Smartphone | 800   | 20       | 16000       |
| 2 | Tablet     | 300   | 15       | 4500        |

--- Top Product by Inventory Value ---

|                        |            |
|------------------------|------------|
| Product                | Smartphone |
| Price                  | 800        |
| Quantity               | 20         |
| Total_Value            | 16000      |
| Name: 1, dtype: object |            |

Total Inventory Value per Product





